

Ujaan Das

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EDUCATION

University of Warwick

Master of Science in Computer Science (Predicted Distinction)

Hong Kong University of Science and Technology

Bachelor of Engineering in Computer Engineering

Coventry, United Kingdom

Sep. 2025 – Sep. 2026

Hong Kong SAR

Sep. 2021 – May 2025

EXPERIENCE

Stellarus Technology

Software Engineering Intern (Part Time)

Jan. 2025 – Aug. 2025

Hong Kong SAR

- Re-designed a chatbot execution pipeline in Python using a multi-step LLM workflow, implementing a custom command DSL for structured tool orchestration that reduced token overhead by 25% per session
- Built a reusable React chat widget with multi-session support, enabling concurrent chats on platform
- Integrated into SpaceGPT and awarded a Gold Medal at the 50th Geneva International Invention Exhibition

Research Assistant

CASTLE Lab, HKUST

May 2024 – Sep. 2024

Hong Kong SAR

- Built a scalable Java static-analysis platform using Spoon to analyze JUnit test suites across 30k+ repositories
- Developed graph-based analysis pipelines to model Java code structure, dependency relationships, control flow, and API interactions across multi-million-line codebases, generating minimally runnable unit test contexts
- Automated build workflows for Maven and Gradle projects using Nix, generating over 35k usable examples

Stellarus Technology

Software Engineering Intern

May 2023 – Aug. 2023

Hong Kong SAR

- Developed RESTful APIs using FastAPI and MongoDB for multi-user task queues for live satellite data queries
- Designed to sustain throughput of 8,000+ requests/sec with sub-150ms latency and minimal queue times
- Parallelized CPU-bottlenecked operations via a Celery/Redis worker pool, reducing processing time by 34%
- Partnered with frontend team to containerize the SPA and backend using Docker and orchestrated automated CI/CD pipelines for Azure Container Apps, cutting deployment time by 20%

HCI Initiative, HKUST

Research Assistant

Jun. 2022 – Aug. 2022

Hong Kong SAR

- Developed a React/TypeScript SPA utilizing Custom Hooks and Context API for centralized state handling
- Engineered a real-time telemetry and metrics pipeline processing 50K+ events/session, enabling analysis across 4K+ user interactions in large-scale experiments
- Worked with other post-docs to build a standardized TailwindCSS and MaterialUI component library
- Contributed to FARPLS, a feature-augmented preference learning system (published at ACM/IUT'24)

PROJECTS

MSc Dissertation - Invariants | C++, Google Test Suite, CMake, Python

[Github](#)

- Applying formal systems theory to LLMs for improved structured output generation via runtime constraints
- Architecting a C++ DSL and compiler with Python bindings for seamless integration into inference pipelines

Snip | C++, Termios, Google Test Suite, CMake, Nix

[Github](#)

- Actively maintaining a modal text editor built from ground-up with asynchronous, event-driven architecture

SKILLS

Languages: C++, Python, Go, C, Java, TypeScript, HTML/CSS, SQL

Frameworks & Libraries: Google Test Suite, CMake, JUnit, Maven, FastAPI, Next.js, React.js, Node.js

Databases: MongoDB, PostgreSQL, MySQL, SQLite, Redis

Tools & Cloud: Docker, Nix, Azure, AWS, GitHub Actions, NGINX, Make, Git

AWARDS

HackEurope'26 Scholar, Paris | *Selected for full scholarship by SIG, Anthropic, and Warwick*

Mar. 2026

Track Runner Up, Imperial College London ICHack | *Tackled HRT's Big Data challenge*

Feb. 2026

Global Merit Scholar | *Competitive scholarship for study abroad at Northwestern University, USA*

2024

Track Winner, Cathay Pacific Hackathon | *Ranked 1st out of 1,000+ global applicants (Top 0.01%)*

Nov. 2023

Published Researcher, ACM IUT'24 | *Paper on human-AI interaction*

Sep. 2023